Edward Silva

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Education

Colorado School of Mines May 2026

BS Electrical Engineering, Minor in Computer Science, GPA: 3.44/4.00 Golden, CO

Honors: Dean's List (Fall 2022, Fall 2024, Spring 2026) | Honor Roll (Fall 2023, Spring 2024, Fall 2025)

Scholarships: Provost Scholarship | Computing-Mines Affiliates Partnership Program (C-MAPP) Scholar, 2023–2025 (Frontier Technology Inc., Parsons Corporation) | American Bureau of Shipping Scholar, 2024–2025

Graduate coursework: Estimation Theory & Kalman Filtering (A)

Relevant coursework: Digital Signal Processing, Modern Control Design, Feedback Control Systems, Embedded Systems, Engineering Electromagnetics, Elements of Computing Systems, Printed Circuit Board Design

Skills

Languages: C, C++, Python, MATLAB
Tools: Simulink, Git, Linux, Intel VTune, OpenCV, Arduino, Autodesk Revit, AutoCAD MEP
Methods: Kalman filtering (EKF, UKF), LQR and observer design, SIMD (AVX-512), FIR filters, I2C

Experience

Embedded Software Engineer, SpaceX Jul 2026 – Present
C, C++, Linux, Git Bastrop, TX

- Write embedded C and C++ software and Linux kernel drivers for Starlink phased array systems, interfacing with the RF, silicon, and network software teams
- Cut the run time of a calculation in the Starlink phased array software by 11%

Software Engineering Intern, Kratos Defense Jun – Aug 2025
C++, Python, SIMD, Git Colorado Springs, CO

- Sped up legacy real-time C++ DSP code by 1.6x with refactoring and AVX-512 vectorization, guided by VTune
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a configurable logging framework with severity levels and error tracking for the real-time DSP software

Co-op Intern, Electrical Design, Jordan and Skala Engineers Jan – Jun 2025
Autodesk Revit, AutoCAD MEP, NEC Compliance Denver, CO

- Drafted layouts, panel schedules, and riser diagrams for 20+ apartment and early data center projects, with NEC load and voltage-drop calculations

Teaching

Calculus II and Differential Equations Grader Jan – Aug 2024
Colorado School of Mines Remote

- Graded Calculus II and Differential Equations for 30–50 students a semester across 1–2 classes

Peer Mentor Aug – Dec 2023
Colorado School of Mines Golden, CO

- Co-taught a 30-student section of the one-credit First Year Success Seminar and led an orientation group

Projects

Electromagnetic Bicycle Damper (EMBR), MATLAB, Python — Fall 2025 – Spring 2026

- Owned the electromagnet design and calculation verification within an 8-person interdisciplinary team, sponsored by ARUS MR Tech, that retrofitted a production front suspension fork with a semi-active MR damper
- Modeled a ~250-turn, 2 A coil in Python by summing Biot-Savart loop fields with complete elliptic integrals, estimating a 0.15 T peak across the 2.5 mm radial gap modeled at the time
- Designed the flux path through the steel parts so the field crosses the annular gap perpendicular to the fluid flow
- Confirmed in testing that the electromagnet worked as designed and noticeably changed the damping

Nonlinear State Estimation for the EMBR Damper (graduate course, solo), MATLAB, Simulink — Spring 2026

- Modeled the MR damper, MOSFET-driven coil, and sensor array as a 4-state nonlinear system in MATLAB and Simulink, adding terrain force to the state as a random walk
- Implemented and compared extended (EKF) and unscented (UKF) Kalman filters to estimate unmeasured states
- Showed the UKF cut RMS force error from 624 N to 401 N at a larger yield constant, keeping 100% of samples inside 3σ against 58% for the EKF

Pitch-Tracking Systems in Auto-Tune, MATLAB — Spring 2026

- Pair-programmed real-time FFT and autocorrelation (ACF) pitch detectors in MATLAB with a teammate, and found in a 50 Hz to 4 kHz chirp sweep that the ACF failed near 1600 Hz and the FFT near 1800 Hz
- Measured mean per-frame latency of 0.22 ms for the FFT against 0.59 ms for the ACF (2.68x), with 0.26 ms against 0.74 ms standard deviation

Gantry Crane Control System, MATLAB, Simulink — Fall 2025

- Designed a 50 Hz discrete-time gantry-crane controller in the z-domain in a 3-person team, with 0% overshoot, 112.6 s settling against a 120 s spec, and cart velocity ≤ 1 m/s
- Found a 48° phase margin and 0.52 s delay tolerance, confirmed by injecting a 25-sample delay
- Pair-programmed an LQR controller with a full-order observer, comparing reference scaling (3.95 s, 9.02% overshoot) with integral augmentation plus anti-windup (3.45 s, negligible overshoot)

Autonomous Maze-Navigating Robot, Python — Fall 2025

- Achieved the second-fastest course completion time in the cohort with an autonomous maze-navigating robot built by a 4-person team, integrating computer vision, embedded control, and I2C communication
- Built the Python and OpenCV ArUco vision subsystem with a teammate, splitting its 640x480 pipeline into capture, processing, display, and LCD threads so the Arduino stopped waiting on the vision code
- Fixed an I2C packet mismatch by repacking 8-byte Python doubles as 4-byte little-endian floats for the Arduino

eeprep (eeprep.com) — Ongoing

- Wrote a free CC BY 4.0 job guide for EECS students and new graduates, with 30 pages and 340+ linked sources

Certifications

MATLAB Machine Learning Techniques, MathWorks

Organizations

IEEE-HKN, Nu Pi Chapter, Colorado School of Mines, Member — 2026 – Present
IEEE, Member — 2025 – Present
IEEE Young Professionals, Member — 2025 – Present